```
root@DESKTOP-TI2KOT3:~# mkdir 11249a108oslab
root@DESKTOP-TI2KOT3:~# cd 11249a108oslab
root@DESKTOP-TI2KOT3:~/11249a108oslab# pwd
/root/11249a108oslab
root@DESKTOP-TI2KOT3:~/11249a108oslab# cd
root@DESKTOP-TI2KOT3:~# pwd
/root
root@DESKTOP-TI2KOT3:~# ls
11249a108  11249a108oslab  shashank
root@DESKTOP-TI2KOT3:~# rmdir
rmdir: missing operand
Try 'rmdir --help' for more information.
root@DESKTOP-TI2KOT3:~# rmdir  oslab
rmdir: failed to remove 'oslab': No such file or directory
root@DESKTOP-TI2KOT3:~# cd 11249a108oslab
root@DESKTOP-TI2KOT3:~/11249a108oslab# touch a.txt
root@DESKTOP-TI2KOT3:~/11249a108oslab# cp a.txt b.txt
root@DESKTOP-TI2KOT3:~/11249a108oslab# mv a.txt 1.txt
root@DESKTOP-TI2KOT3:~/11249a108oslab# rm 1.txt
root@DESKTOP-TI2KOT3:~/11249a108oslab# ls
b.txt
root@DESKTOP-TI2KOT3:~/11249a108oslab# file b.txt
b.txt: empty
root@DESKTOP-TI2KOT3:~/11249a108oslab# .stat b.txt
Command '.stat' not found, did you mean:
  command 'jstat' from deb openjdk-17-jdk-headless (17.0.12+7-1ubuntu2~24.04)
  command 'jstat' from deb openjdk-21-jdk-headless (21.0.4+7-1ubuntu2~24.04)
  command 'nstat' from deb iproute2 (6.1.0-1ubuntu2)
  command 'stat' from deb coreutils (9.4-2ubuntu2)
Try: apt install <deb name>
root@DESKTOP-TI2KOT3:~/11249a108oslab# stat b.txt
  File: b.txt
  Size: 0               Blocks: 0          IO Block: 4096   regular empty file
Device: 8,48    Inode: 49294       Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-04-24 09:29:13.708163106 +0000
Modify: 2026-04-24 09:29:13.708163106 +0000
Change: 2026-04-24 09:29:13.708163106 +0000
 Birth: 2026-04-24 09:29:13.708163106 +0000
root@DESKTOP-TI2KOT3:~/11249a108oslab# top
top - 09:37:47 up 34 min,  1 user,  load average: 0.03, 0.05, 0.00
```

```
top - 09:38:59 up 35 min,  1 user,  load average: 0.26, 0.10, 0.02
Tasks:  22 total,   1 running,  21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7784.2 total,   7009.2 free,    537.8 used,    388.8 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7246.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   21692  12892   9436 S   0.7   0.2   0:03.72 systemd
     53 root      19  -1   34028  14336  13568 S   0.3   0.2   0:01.06 systemd-journal
      2 root      20   0    3120   1920   1920 S   0.0   0.0   0:00.00 init-systemd(Ub
      7 root      20   0    3136   1796   1792 S   0.0   0.0   0:00.00 init
    105 root      20   0   24872   6016   4864 S   0.0   0.1   0:00.21 systemd-udevd
    112 systemd+  20   0   21452  11776   9856 S   0.0   0.1   0:00.09 systemd-resolve
    113 systemd+  20   0   91020   7680   6784 S   0.0   0.1   0:00.07 systemd-timesyn
    176 root      20   0    4236   2432   2304 S   0.0   0.0   0:00.00 cron
    177 message+  20   0    9520   4864   4224 S   0.0   0.1   0:00.64 dbus-daemon
    184 root      20   0   17956   8448   7552 S   0.0   0.1   0:00.25 systemd-logind
    195 root      20   0    3160   1920   1792 S   0.0   0.0   0:00.01 agetty
    209 syslog    20   0  222508   5248   4480 S   0.0   0.1   0:00.42 rsyslogd
    212 root      20   0    3116   1792   1664 S   0.0   0.0   0:00.00 agetty
    213 root      20   0  107008  22272  13056 S   0.0   0.3   0:00.07 unattended-upgr
    289 root      20   0    3124    896    768 S   0.0   0.0   0:00.00 SessionLeader
    290 root      20   0    3140   1156   1024 S   0.0   0.0   0:00.04 Relay(291)
    291 root      20   0    6068   5120   3456 S   0.0   0.1   0:00.05 bash
    292 root      20   0    6660   4096   3456 S   0.0   0.1   0:00.00 login
    335 root      20   0   20304  11392   9344 S   0.0   0.1   0:00.05 systemd
    336 root      20   0   21148   3500   1792 S   0.0   0.0   0:00.00 (sd-pam)
    358 root      20   0    6072   5120   3584 S   0.0   0.1   0:00.01 bash
   7008 root      20   0    9292   5504   3328 R   0.0   0.1   0:00.03 top
```

```
  0[                          0.0%]   4[                          0.0%]   8[                          0.0%] 12[|                         0.7%]
  1[                          0.0%]   5[                          0.0%]   9[                          0.0%] 13[                         0.0%]
  2[                          0.0%]   6[                          0.0%]  10[                          0.0%] 14[                         0.0%]
  3[                          0.0%]   7[                          0.0%]  11[                          0.0%] 15[                         0.0%]
Mem[||||||||||                                     391M/7.60G] Tasks: 24, 13 thr, 0 kthr; 1 running
Swp[                                                 0K/2.00G] Load average: 0.19 0.09 0.02
                                                               Uptime: 00:41:51

  Main   I/O
    PID USER      PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
      1 root       20   0 21692 12892  9436 S   0.0  0.2  0:04.71 /sbin/init
      2 root       20   0  3120  1920  1920 S   0.0  0.0  0:00.00 /init
      7 root       20   0  3136  1796  1792 S   0.0  0.0  0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
      8 root       20   0  3136  1796  1792 S   0.0  0.0  0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
      9 root       20   0  3120  1920  1920 S   0.0  0.0  0:00.00 /init
     53 root       19  -1 34028 14976 14208 S   0.0  0.2  0:01.36 /usr/lib/systemd/systemd-journald
    105 root       20   0 24872  6016  4864 S   0.0  0.1  0:00.25 /usr/lib/systemd/systemd-udevd
    112 systemd-re 20   0 21452 11776  9856 S   0.0  0.1  0:00.09 /usr/lib/systemd/systemd-resolved
    113 systemd-ti 20   0 91020  7680  6784 S   0.0  0.1  0:00.07 /usr/lib/systemd/systemd-timesyncd
    137 systemd-ti 20   0 91020  7680  6784 S   0.0  0.1  0:00.00 /usr/lib/systemd/systemd-timesyncd
    176 root       20   0  4236  2432  2304 S   0.0  0.0  0:00.00 /usr/sbin/cron -f -P
    177 messagebus 20   0  9652  4864  4224 S   0.0  0.1  0:00.82 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
    184 root       20   0 17956  8448  7552 S   0.0  0.1  0:00.31 /usr/lib/systemd/systemd-logind
    195 root       20   0  3160  1920  1792 S   0.0  0.0  0:00.01 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
    209 syslog     20   0  217M  5248  4480 S   0.0  0.1  0:00.02 /usr/sbin/rsyslogd -n -iNONE
    212 root       20   0  3116  1792  1664 S   0.0  0.0  0:00.00 /sbin/agetty -o -p -- \u --noclear - linux
    213 root       20   0  104M 22272 13056 S   0.0  0.3  0:00.07 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
    219 syslog     20   0  217M  5248  4480 S   0.0  0.1  0:00.36 /usr/sbin/rsyslogd -n -iNONE
    220 syslog     20   0  217M  5248  4480 S   0.0  0.1  0:00.00 /usr/sbin/rsyslogd -n -iNONE
    221 syslog     20   0  217M  5248  4480 S   0.0  0.1  0:00.13 /usr/sbin/rsyslogd -n -iNONE
    254 root       20   0  104M 22272 13056 S   0.0  0.3  0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
    289 root       20   0  3124   896   768 S   0.0  0.0  0:00.00 /init
    290 root       20   0  3140  1156  1024 S   0.0  0.0  0:00.06 /init
    291 root       20   0  6068  5120  3456 S   0.0  0.1  0:00.06 -bash
    292 root       20   0  6660  4096  3456 S   0.0  0.1  0:00.00 /bin/login -f
    335 root       20   0 20304 11392  9344 S   0.0  0.1  0:00.05 /usr/lib/systemd/systemd --user
    336 root       20   0 21148  3500  1792 S   0.0  0.0  0:00.00 (sd-pam)
    358 root       20   0  6072  5120  3584 S   0.0  0.1  0:00.01 -bash
   9490 root       20   0  361M 20224 17408 S   0.0  0.3  0:00.03 /usr/libexec/packagekitd
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

```
root@DESKTOP-TI2KOT3:~/11249a108oslab# htop
root@DESKTOP-TI2KOT3:~/11249a108oslab# kill 9
root@DESKTOP-TI2KOT3:~/11249a108oslab# pkill root
root@DESKTOP-TI2KOT3:~/11249a108oslab# jobs
root@DESKTOP-TI2KOT3:~/11249a108oslab# fg
-bash: fg: current: no such job
root@DESKTOP-TI2KOT3:~/11249a108oslab# bg
-bash: bg: current: no such job
root@DESKTOP-TI2KOT3:~/11249a108oslab# uptime
 09:47:44 up 44 min,  1 user,  load average: 0.01, 0.05, 0.00
root@DESKTOP-TI2KOT3:~/11249a108oslab# who
root     pts/1        2026-04-24 09:16
root@DESKTOP-TI2KOT3:~/11249a108oslab# whoami
root
root@DESKTOP-TI2KOT3:~/11249a108oslab# free
              total        used        free      shared  buff/cache   available
Mem:        7971004      562332     7106188        3612      460144     7408672
Swap:       2097152           0     2097152
root@DESKTOP-TI2KOT3:~/11249a108oslab# df
Filesystem      1K-blocks      Used  Available Use% Mounted on
none              3985500         0    3985500   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none              3985500         4    3985496   1% /mnt/wsl
drivers         501602300 169509712  332092588  34% /usr/lib/wsl/drivers
/dev/sdd       1055762868   1222848 1000836548   1% /
none              3985500        76    3985424   1% /mnt/wslg
none              3985500         0    3985500   0% /usr/lib/wsl/lib
rootfs            3980460      2720    3977740   1% /init
none              3985500       548    3984952   1% /run
none              3985500         0    3985500   0% /run/lock
none              3985500         0    3985500   0% /run/shm
none              3985500        76    3985424   1% /mnt/wslg/versions.txt
none              3985500        76    3985424   1% /mnt/wslg/doc
C:\             501602300 169509712  332092588  34% /mnt/c
D:\             495660028  53736880  441923148  11% /mnt/d
tmpfs              797100        20     797080   1% /run/user/0
root@DESKTOP-TI2KOT3:~/11249a108oslab#
```

```
root@DESKTOP-TI2KOT3:~/11249a108oslab# du
4       .
root@DESKTOP-TI2KOT3:~/11249a108oslab# cat>b.txt
i am shashank
hello
^C
root@DESKTOP-TI2KOT3:~/11249a108oslab# cat>b.txt
hiii
iam shashank
i am doing operating system lab
^C
root@DESKTOP-TI2KOT3:~/11249a108oslab# cat b.txt
hiii
iam shashank
i am doing operating system lab
root@DESKTOP-TI2KOT3:~/11249a108oslab# tac b.txt
i am doing operating system lab
iam shashank
hiii
root@DESKTOP-TI2KOT3:~/11249a108oslab# more b.txt
hiii
iam shashank
i am doing operating system lab
root@DESKTOP-TI2KOT3:~/11249a108oslab# less b.txt
root@DESKTOP-TI2KOT3:~/11249a108oslab# head b.txt
hiii
iam shashank
i am doing operating system lab
root@DESKTOP-TI2KOT3:~/11249a108oslab# tail b.txt
hiii
iam shashank
i am doing operating system lab
root@DESKTOP-TI2KOT3:~/11249a108oslab# grep "shashank" b.txt
iam shashank
root@DESKTOP-TI2KOT3:~/11249a108oslab# c b.txt
c: command not found
root@DESKTOP-TI2KOT3:~/11249a108oslab# wc b.txt
 3  9 50 b.txt
root@DESKTOP-TI2KOT3:~/11249a108oslab# sort b.txt
hiii
```

```
root@DESKTOP-TI2KOT3:~/11249a108oslab# sort b.txt
hiii
i am doing operating system lab
iam shashank
root@DESKTOP-TI2KOT3:~/11249a108oslab# uniq b.txt
hiii
iam shashank
i am doing operating system lab
root@DESKTOP-TI2KOT3:~/11249a108oslab# cut -d':' -f1 /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
systemd-timesync
dhcpcd
messagebus
syslog
systemd-resolve
uuidd
landscape
polkitd
```

```
root@DESKTOP-TI2KOT3:~/11249a108oslab# touch a.txt
root@DESKTOP-TI2KOT3:~/11249a108oslab# paste b.txt a.txt
hiii
iam shashank
i am doing operating system lab
root@DESKTOP-TI2KOT3:~/11249a108oslab# tr a-z A-Z<b.txt
HIII
IAM SHASHANK
I AM DOING OPERATING SYSTEM LAB
root@DESKTOP-TI2KOT3:~/11249a108oslab# vi 3.py
root@DESKTOP-TI2KOT3:~/11249a108oslab# python3 3.py
Traceback (most recent call last):
  File "/root/11249a108oslab/3.py", line 1, in <module>
    helllooooo
    ^^^^^^^^^^
NameError: name 'helllooooo' is not defined
root@DESKTOP-TI2KOT3:~/11249a108oslab# nano b.txt
root@DESKTOP-TI2KOT3:~/11249a108oslab# ls -l
total 8
-rw-r--r-- 1 root root 20 Apr 24 10:06 3.py
-rw-r--r-- 1 root root  0 Apr 24 10:05 a.txt
-rw-r--r-- 1 root root 57 Apr 24 10:10 b.txt
root@DESKTOP-TI2KOT3:~/11249a108oslab# touch 6.sh
root@DESKTOP-TI2KOT3:~/11249a108oslab# chmod u+x 6.sh
root@DESKTOP-TI2KOT3:~/11249a108oslab# chmod shashank 2.txt
chmod: invalid mode: ‘shashank’
Try 'chmod --help' for more information.
root@DESKTOP-TI2KOT3:~/11249a108oslab# chown shashanka.txt
chown: missing operand after ‘shashanka.txt’
Try 'chown --help' for more information.
root@DESKTOP-TI2KOT3:~/11249a108oslab#
```